

INSTITUTO TECNOLÓGICO DE SALTILLO

Computo en la Nube

Actividad 5.8: Instalación de entorno LAMP

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- Acceder a portal.azure.com

The screenshot shows the Azure portal home page. At the top, there's a section titled "Azure services" with icons for "Create a resource", "Virtual machines", "Subscription", "Resource groups", "Cost Management", "Storage accounts", "Security", "SQL server", "App Services", and "More services". Below this is a "Resources" section with tabs for "Recent" and "Favorite". The "Recent" tab is active, showing a table of resources:

Name	Type	Last Viewed
Azure for Students	Subscription	5 days ago
Computo-Nube	Resource group	5 days ago

Below the table is a "See all" link. Further down is a "Navigate" section with icons for "Subscriptions", "Resource groups", "All resources", and "Dashboard". At the bottom is a "Tools" section with four tiles: "Microsoft Learn" (Learn Azure with free online training from Microsoft), "Azure Monitor" (Monitor your apps and infrastructure), "Microsoft Defender for Cloud" (Secure your apps and infrastructure), and "Cost Management" (Analyze and optimize your cloud spend for free).

- Crear grupo de recursos

The screenshot shows the Azure portal page for the "Computo-Nube" resource group. The page has a left sidebar with navigation links: "Overview", "Activity log", "Access control (IAM)", "Tags", "Resource visualizer", "Events", "Settings", "Cost Management", "Monitoring", "Automation", and "Help". The main content area is titled "Computo-Nube" and includes a search bar and action buttons: "+ Create", "Manage view", "Delete resource group", "Refresh", "Export to CSV", "Open query", "Assign tags", "Move", and "Delete". Below these are "Essentials" and "Resources" sections. The "Essentials" section shows:

- Subscription (move): [Azure for Students](#)
- Subscription ID: f695d551-013c-4914-bb47-02adcc961c45
- Tags (edit): [Add tags](#)
- Deployments: No deployments
- Location: East US

The "Resources" section has a filter bar with "Filter for any f...", "Type equals all", "Location equals all", and an "Add filter" button.

- Crear VM Ubuntu Server

Home > Compute infrastructure | Virtual machines

Create a virtual machine

folders to organize and manage all your resources.

Subscription * ⓘ Azure for Students

Resource group * ⓘ Computo-Nube
[Create new](#)

Instance details

Virtual machine name * ⓘ AppServer ✓

Region * ⓘ (US) East US
[Deploy to an Azure Extended Zone](#)

Availability options ⓘ Availability zone

Zone options ⓘ

- ☒ Self-selected zone
Choose up to 3 availability zones, one VM per zone
- ☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

i Using an Azure-selected zone is not supported in region 'East US'.

Availability zone * ⓘ Zone 1

✔ You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type ⓘ Trusted launch virtual machines
[Configure security features](#)

Image * ⓘ Ubuntu Server 24.04 LTS - x64 Gen2
[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

- ☐ Arm64
- ☒ x64

< Previous **Next : Disks >** **Review + create**

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Microsoft

Search

Virtual machines Get started

+ Create Reservations Manage view Refresh Export to CSV Open query Assign tags

Filter for any f... Subscription equals all Type equals all Resource Group equals all Location equals all

	Name ↑	Subscription	Resource Group	Location	Status	Operating
☐	AppServer	Azure for Stu...	Computo-Nube	East US	Running	Linux

> Disks + images
> Capacity + placement

AppServer
Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search Help me copy this VM in any region

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
- Networking
- Settings
- Availability + scale
- Security
- Backup + disaster recovery
- Operations
- Monitoring
- Automation
- Help

Essentials [JSON View](#)

Resource group ([move](#))
[Compute-Nube](#)

Status
Running

Location
East US (Zone 1)

Subscription ([move](#))
[Azure for Students](#)

Subscription ID
f695d551-013c-4914-bb47-02adcc961c45

Availability zone
1

Operating system
Linux (ubuntu 24.04)

Size
Standard FX8-2mds v2 (2 vcpus, 168 GiB memory)

Primary NIC public IP
[172.203.218.20](#)
[1 associated public IPs](#)

Virtual network/subnet
[AppServer-vnet/default](#)

DNS name
[Not configured](#)

Health state
-

Time created
4/30/2026, 3:10 AM UTC

Tags ([edit](#))
[Add tags](#)

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name	AppServer
Operating system	Linux (ubuntu 24.04)
VM generation	V2
VM architecture	x64
Agent status	Ready
Agent version	2.15.1.3
Hibernation	Disabled

Networking

Public IP address	172.203.218.20 (Network interface appserver606_z1) 1 associated public IPs
Public IP address - (IPv6)	-
Private IP address	10.0.0.4
Private IP	-

Add or remove favorites by pressing

- Abrir puertos 22 y 80

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload Help me choose the right VM size for my workload

Administrator account

Authentication type

☒ SSH public key
☐ Password

Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username *

azureuser ✓

SSH public key source

Generate new key pair ✓

SSH Key Type

☒ RSA SSH Format
☐ Ed25519 SSH Format

Ed25519 provides a fixed security level of no more than 128 bits for 256-bit key, while RSA could offer better security with keys longer than 3072 bits.

Key pair name *

AppServer_key ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

☐ None
☒ Allow selected ports

Select inbound ports *

HTTP (80), SSH (22) ✓

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous Next : Disks > Review + create

- Conectarse por SSH

```
~  
$ sudo chmod 400 .ssh/AppServer_key.pem  
[sudo] password for tarmairon:
```

Se crea config de ssh para tener mas facilidad al conectarse

```
1 Host AppServer
2   HostName 172.203.218.20
3   User azureuser
4   IdentityFile /home/tarmairon/.ssh/AppServer_key.pem
5
```

```
> ssh AppServer
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Apr 30 03:22:42 UTC 2026

System load:  0.08                Temperature:   71.8 C
Usage of /:   5.7% of 28.02GB     Processes:    118
Memory usage: 0%                Users logged in: 0
Swap usage:   0%                IPv4 address for eth0: 10.0.0.4

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@AppServer:~$
```

- Actualizar sistema

```

azureuser@AppServer:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1920 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [346 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.1 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1685 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [324 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.5 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3054 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [702 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [480 B]
Get:25 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [44.4 kB]
Get:26 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [10.2 kB]
Get:27 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:28 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [656 B]
Get:29 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.6 kB]
Get:30 http://azure.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172 B]
Get:31 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7364 B]
Get:32 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:33 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.6 kB]

```

```

azureuser@AppServer:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  gir1.2-packagekitglib-1.0 iproute2 jq libcap2 libcap2-bin libjq1 libntfs-3g89t64 libpackagekit-glib2-18 libpam-cap
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 ntfs-3g openssh-client openssh-server openssh-sftp-server packagekit
  packagekit-tools python3.12 python3.12-minimal snapd ubuntu-pro-client ubuntu-pro-client-l10n vim vim-common vim-runtime vim-tiny xxd
28 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
24 standard LTS security updates
Need to get 58.2 MB of archives.
After this operation, 1434 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12t64 amd64 3.12.3-1ubuntu0.13 [2338 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12 amd64 3.12.3-1ubuntu0.13 [662 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-stdlib amd64 3.12.3-1ubuntu0.13 [2068 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12-minimal amd64 3.12.3-1ubuntu0.13 [2346 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-minimal amd64 3.12.3-1ubuntu0.13 [837 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 ntfs-3g amd64 1:2022.10.3-1.2ubuntu3.1 [406 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libntfs-3g89t64 amd64 1:2022.10.3-1.2ubuntu3.1 [159 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.16 [37.3 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.16 [510 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.16 [907 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2 amd64 1:2.66-5ubuntu2.4 [30.5 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpam-cap amd64 1:2.66-5ubuntu2.4 [12.5 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcap2-bin amd64 1:2.66-5ubuntu2.4 [34.1 kB]

```

- Instalar Apache


```

azureuser@AppServer:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1335 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.11 [90.2 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2091 kB in 0s (15.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 69212 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.

```

```

azureuser@AppServer:~$ sudo systemctl status apache2
WARNING: terminal is not fully functional
Press RETURN to continue
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-30 03:25:13 UTC; 27s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 7677 (apache2)
     Tasks: 55 (limit: 202807)
    Memory: 5.3M (peak: 5.5M)
       CPU: 21ms
   CGroup: /system.slice/apache2.service
           └─7677 /usr/sbin/apache2 -k start
             └─7680 /usr/sbin/apache2 -k start
               └─7681 /usr/sbin/apache2 -k start

Apr 30 03:25:13 AppServer systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 30 03:25:13 AppServer systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@AppServer:~$

```

Not Secure http://172.203.218.20



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
├─ apache2.conf
│   └─ ports.conf
├─ mods-enabled
│   └─ *.load
│       └─ *.conf
├─ conf-enabled
│   └─ *.conf
└─ sites-enabled
    └─ *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.

- Instalar PHP

```
azureuser@AppServer:~$ sudo apt install php libapache2-mod-php php-mysql -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libapache2-mod-php8.3 php-common php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache php8.3-readline
Suggested packages:
  php-pear
The following NEW packages will be installed:
  libapache2-mod-php libapache2-mod-php8.3 php php-common php-mysql php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache
  php8.3-readline
0 upgraded, 11 newly installed, 0 to remove and 0 not upgraded.
Need to get 5051 kB of archives.
After this operation, 22.9 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.8 [740 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.8 [371 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.8 [13.5 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cli amd64 8.3.6-0ubuntu0.24.04.8 [1916 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapache2-mod-php8.3 amd64 8.3.6-0ubuntu0.24.04.8 [1850 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libapache2-mod-php all 2:8.3+93ubuntu2 [4224 B]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3 all 8.3.6-0ubuntu0.24.04.8 [9170 B]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php all 2:8.3+93ubuntu2 [4076 B]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-mysql amd64 8.3.6-0ubuntu0.24.04.8 [127 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 php-mysql all 2:8.3+93ubuntu2 [1838 B]
Fetched 5051 kB in 0s (15.4 MB/s)
Selecting previously unselected package php-common.
(Reading database ... 69935 files and directories currently installed.)
Preparing to unpack .../00-php-common_2%3a93ubuntu2_all.deb ...
Unpacking php-common (2:93ubuntu2) ...
Selecting previously unselected package php8.3-common.
Preparing to unpack .../01-php8.3-common_8.3.6-0ubuntu0.24.04.8_amd64.deb ...
Unpacking php8.3-common (8.3.6-0ubuntu0.24.04.8) ...
Selecting previously unselected package php8.3-opcache.
Preparing to unpack .../02-php8.3-opcache_8.3.6-0ubuntu0.24.04.8_amd64.deb ...
Unpacking php8.3-opcache (8.3.6-0ubuntu0.24.04.8) ...
Selecting previously unselected package php8.3-readline.
Preparing to unpack .../03-php8.3-readline_8.3.6-0ubuntu0.24.04.8_amd64.deb ...
Unpacking php8.3-readline (8.3.6-0ubuntu0.24.04.8) ...
```

```
azureuser@AppServer:~$ php -v
PHP 8.3.6 (cli) (built: Mar 20 2026 02:32:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
azureuser@AppServer:~$
```

```
DirectoryIndex index.php index.html index.cgi index.p index.xhtml index.htm
~
~
~
~
~
~
```

```
~
"/etc/apache2/mods-enabled/dir.conf" 1L, 77B written
azureuser@AppServer:~$ sudo systemctl restart apache2
azureuser@AppServer:~$
```

```
~
~
~
"/var/www/html/info.php" 3L, 20B written
azureuser@AppServer:~$ cat /var/www/html/info.php
<?php
phpinfo();
?>
azureuser@AppServer:~$
```

PHP Version 8.3.6



System	Linux AppServer 6.17.0-1011-azure #11~24.04.2-Ubuntu SMP Wed Mar 25 22:46:36 UTC 2026 x86_64
Build Date	Mar 20 2026 02:32:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-mysqld.ini, /etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-ffi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-mysqli.ini, /etc/php/8.3/apache2/conf.d/20-pdo_mysql.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.3/apache2/conf.d/20-sysvsem.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831,NTS
PHP Extension Build	API20230831,NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled
Zend Multibyte Support	disabled
Zend Max Execution Timers	disabled
IPv6 Support	enabled
DTrace Support	disabled
Registered PHP Streams	https, ftps, compress.zlib, php, file, glob, data, http, ftp, phar
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, tls, tlsv1.0, tlsv1.1, tlsv1.2, tlsv1.3
Registered Stream Filters	zlib.*, string.rot13, string.toupper, string.tolower, convert.*, consumed, dechunk, convert.iconv.*

- Instalar MySQL


```
azureuser@AppServer:~$ sudo apt install mariadb-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 libfcgi-fast-perl libfcgi-pm-perl libclone-perl libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libencode-locale-perl
  libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21 libsnappy1v5 libtimedate-perl liburi-perl
  liburing2 mariadb-client mariadb-client-core mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
  mariadb-plugin-provider-lzma mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server-core mysql-common pv socat
Suggested packages:
  libmldbm-perl libnet-daemon-perl libsql-statement-perl libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl
  libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx mariadb-test doc-base
The following NEW packages will be installed:
  galera-4 libfcgi-fast-perl libfcgi-pm-perl libclone-perl libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libencode-locale-perl
  libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmariadb3 libmysqlclient21 libsnappy1v5 libtimedate-perl liburi-perl
  liburing2 mariadb-client mariadb-client-core mariadb-common mariadb-plugin-provider-bzip2 mariadb-plugin-provider-lz4
  mariadb-plugin-provider-lzma mariadb-plugin-provider-lzo mariadb-plugin-provider-snappy mariadb-server mariadb-server-core mysql-common
  pv socat
0 upgraded, 37 newly installed, 0 to remove and 0 not upgraded.
Need to get 19.7 MB of archives.
After this operation, 201 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 galera-4 amd64 26.4.16-2build4 [736 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-common all 1:10.11.14-0ubuntu0.24.04.1 [28.6 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libdbi-perl amd64 1.643-4build3 [721 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libconfig-inifiles-perl all 3.000003-2 [39.4 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 libmariadb3 amd64 1:10.11.14-0ubuntu0.24.04.1 [196 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-client-core amd64 1:10.11.14-0ubuntu0.24.04.1 [1039 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-client amd64 1:10.11.14-0ubuntu0.24.04.1 [2502 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 liburing2 amd64 2.5-1build1 [21.1 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-server-core amd64 1:10.11.14-0ubuntu0.24.04.1 [8406 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 mariadb-server amd64 1:10.11.14-0ubuntu0.24.04.1 [237 kB]
```

```
azureuser@AppServer:~$ sudo mariadb-secure-installation
```

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):

OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y

Enabled successfully!

Reloading privilege tables..

... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y

New password:

Re-enter new password:

Password updated successfully!

Reloading privilege tables..

... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y

... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y

... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y

- Dropping test database...

... Success!

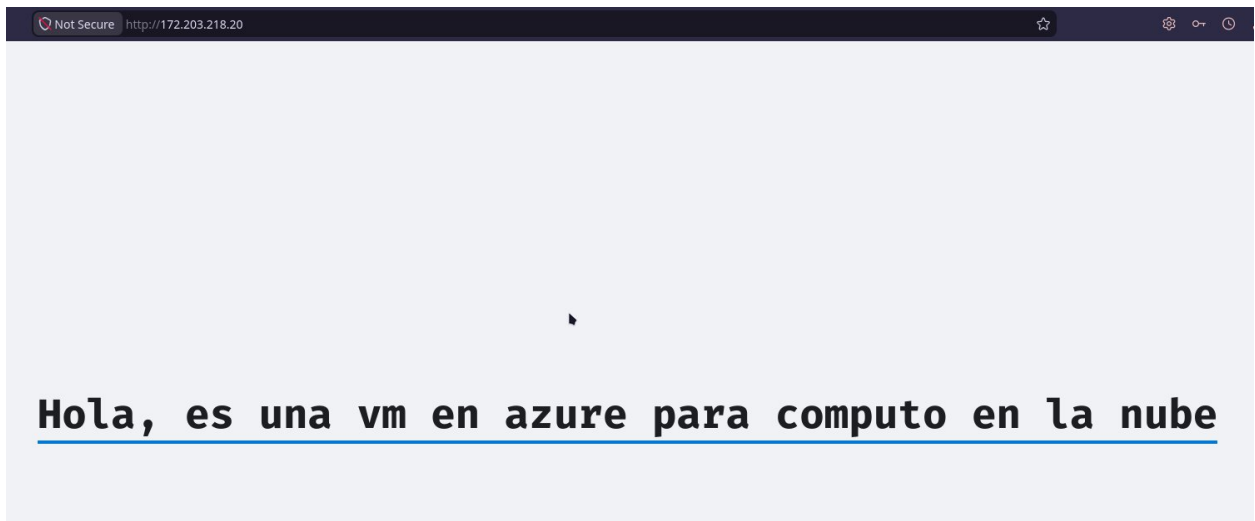
- Removing privileges on test database...

```
azureuser@AppServer:~$  
azureuser@AppServer:~$ sudo mysql -u root -p  
Enter password:  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 41  
Server version: 10.11.14-MariaDB-0ubuntu0.24.04.1 Ubuntu 24.04  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
No entry for terminal type "xterm-kitty";  
using dumb terminal settings.  
No entry for terminal type "xterm-kitty";  
using dumb terminal settings.  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]>
```

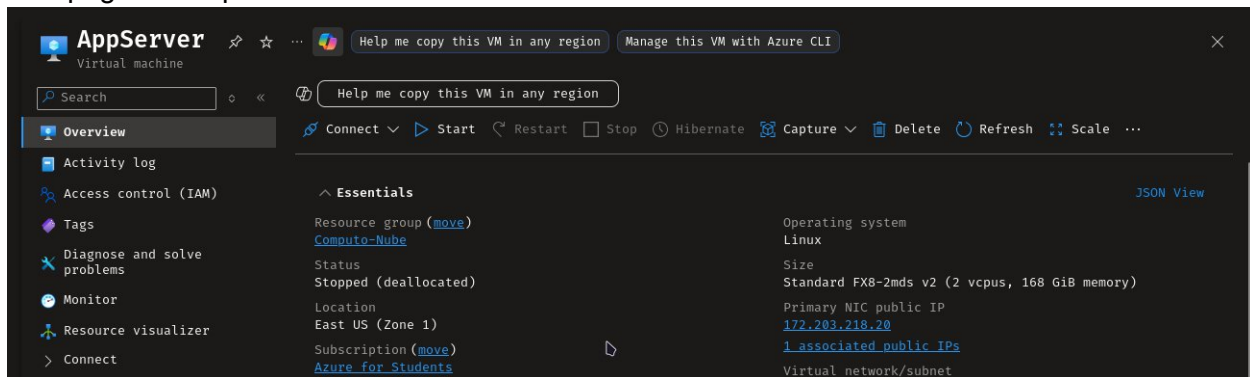
- Probar servidor web

No se que se deba hacer aqui, asi que hice un index.html rapido

```
<html>  
<head>  
  <style>  
    body { font-family: sans-serif; display: flex; justify-content: center; align-items: center; height: 100vh; margin: 0; background-c  
olor: #f0f2f5; }  
    h1 { color: #1c1e21; font-size: 3rem; border-bottom: 4px solid #0078d4; }  
  </style>  
</head>  
<body>  
  <h1>Hola, es una vm en azure para computo en la nube</h1>  
</body>  
</html>
```



Se apaga la maquina



Conclusion

Si ahorra bastante tiempo, hubiera ahorrado una cantidad de tiempo mayor si no me hubiera tenido que pelear con azure sobre imagenes de vm que puedo usar durante 2 dias, al final tuve que poner una tarjeta que no uso, de que sirve la subscripcion de estudiantes? Serviran aun mis creditos? Ni idea, de todos los servicios de nube que he usado este es el peor sin duda incluso sin eso la UI es muy fea. Al menos no tengo que configurar yo las maquinas virtuales de nuevo, nomas las creas y ya sirve todo